TAKE HOME CASE REPORT

RenewFlow

Reengineering and Automating the Student Renewal-to-Class-Scheduling Pipeline

Prepared by	Ankesh, B.Tech Computer Science and Engineering, Lovely Professional University
Prepared for	Bambinos.live, Head of Operations and the Hiring Team
Assessment	Hiring Assessment, Stage 2 (Operations)
Role applied for	AI Program Manager (Operations), Bengaluru
Date of submission	7 November 2026
Submitted with	This report and the RenewFlow project repository (link in the submission form)

2. Executive Summary

Problem. When a parent pays to renew a child's classes, the next term often takes days to appear on the calendar. The renewal passes through four tables (invoice, teacher form, booking request, class schedule) with three manual handoffs, and nothing watches the chain as a whole.

Approach. I read all five CSV files in the official repository, traced each of the 13 invoices by hand, grouped the failures, and then built **RenewFlow**, a FastAPI reconciliation engine with an eight-page web interface. It auto-approves clean renewals, routes exceptions to a named owner with a reason code and timer, and drafts the next WhatsApp or Slack message.

Key findings. At 9:00 AM on 4 November 2026 (the reference time used throughout), only 1 of 13 renewals is scheduled, and it has 2 of 24 sessions on the calendar. Rs 126,850 has been collected on invoices with no schedule. Teachers are not the bottleneck (median 27 minutes from payment to form); the delay builds in the Operations queue and downstream, where 3 of 4 approvals stalled.

Outcome. On the sample, RenewFlow approves 6 of 13 with no human touch, sends 2 to the parent for one tap, flags 3, blocks 1 (failed payment) and leaves 1 alone (complete). It matches my hand-built answer on all 13 invoices and 8 metrics, and 44 automated tests pass. Under stated assumptions (400 renewals a week), Operations time could fall from about 120 to about 20 hours a week. These savings are assumptions to be replaced with real volumes.

Recommendation. Take the teacher form off the critical path, let rules decide money, identity and seats, use AI only for language, and roll out in three gated stages (shadow, assist, autopilot) over 90 days with a kill switch.

3. Introduction

Bambinos.live runs thousands of live small-group online classes each week for children aged 4 to 16 in India, the Gulf, the US and the UK. Renewals drive a large share of revenue and retention: a parent already in a class pays to continue, usually after a pitch from the child's teacher. Every day a paid renewal stays unscheduled is a day of interrupted classes for a family that has already done its part, and of repetitive checking and chasing for Operations. Fixing this pipeline is therefore both a customer-experience and a cost problem, and it is a good test case for how AI and no-code automation should be applied: only where they are safe and checkable.

4. Problem Statement

Design and demonstrate a way to move a renewal from *Payment Completed* to *Classes Scheduled* with most cases requiring no manual work, so that parents see no interruption and Operations spends its time on genuine exceptions. The case asks for four deliverables: **A** a root-cause and leakage audit, **B** a reengineered AI and no-code workflow with smart exception handling, **C** a working prototype that reads the five CSVs and outputs auto-approved entries, flagged exceptions with reason codes and drafted messages, and **D** a dashboard, SOPs and a 30/60/90-day plan.

5. Objectives and Scope

- Quantify where renewals leak and where the time goes, using only the data provided.
- Question every manual step before automating it; remove those that only duplicate data.
- Build a deterministic engine whose every decision is explainable, with AI limited to language tasks.
- Provide an operating model (owners, timers, SOPs) and a safe rollout plan.
- **Out of scope:** sending real messages, writing to live tables, fixing multi-currency billing (noted as F16), and any claim of statistical significance from 13 invoices.

6. Dataset Description

Source: the five CSV files in `ops/data` of the official assessment repository. All timestamps are treated as IST. Names, mobile numbers and emails are masked (for example with asterisks), so they are compared as partial values.

File / table	Rows	Role in the chain	Main fields (as used)
invoices	13	System-created anchor; proof of money	student, user/parent account, class, teacher (educator id), price, discount, net amount, gateway, payment id, status, paid-at
faculty_form_unapproved	12	Teacher renewal form (manual)	invoice id/link, student details, amount paid, sessions, same batch or change, preferred days/time, free-text remark
booking_requests	4	Created when Ops approves a form	status (Pending_Batch_Allocation, Failed_Batch_Full, Awaiting_Parent_Batch_Selection, Completed), batch, timestamps
class_schedule_bookings	2	Puts classes on the calendar	booking request, batch, date, start/end time, type, attendance and cancellation flags
reference (batches and students)	13	Source of truth for identity and a clean renewal	registered parent name/mobile/email, current batch, days, start time, teacher, standard fee (24 sessions), batch size vs maximum of 6

Data types and important features

- **Identifiers** (invoice, student, user, form, booking, gateway payment id): the join keys. Typed ones (form to invoice) are the main error source.
- **Money** (price, discount, net amount, form amount, standard fee): decide payment and discount checks.
- **Timestamps**: drive every age, delay and service-level clock.
- **Categorical** (payment status, booking status, batch choice, days): drive the decision tiers.
- **Free text** (teacher remark, masked names): needs language reading and normalisation.

Data quality: missing, duplicate and unusual values

Issue	Evidence in sample	Handling
Missing values	Literal NULL / empty strings; failed payment (84911) has no gateway payment id; no form for 84906 and 84910	Treated as missing; payment gate requires id and paid-at
Duplicates	Forms 3308 and 3309 for invoice 84908, about 5 minutes apart	Earliest kept, copy suppressed if identical and within 24 h
Key error	Form 3313 carries 84931 instead of 84913 (digits swapped)	Healed by scored matching
Look-alike characters	Two Cyrillic letters in the parent name on invoice 84913	Folded to Latin before comparison
Masked fields	Names, phones, emails partly asterisked	Compared on visible prefix and suffix

7. Data Understanding and Exploration

I anchored on the invoices table because it is the only system-created record, and for each invoice asked: was the money received in full, is there exactly one correct form, does the payer match the registered parent, is the batch available, and how far has the record travelled and how long has it stood still. The chain narrows from **13 to 10 to 4 to 4 to 1**.

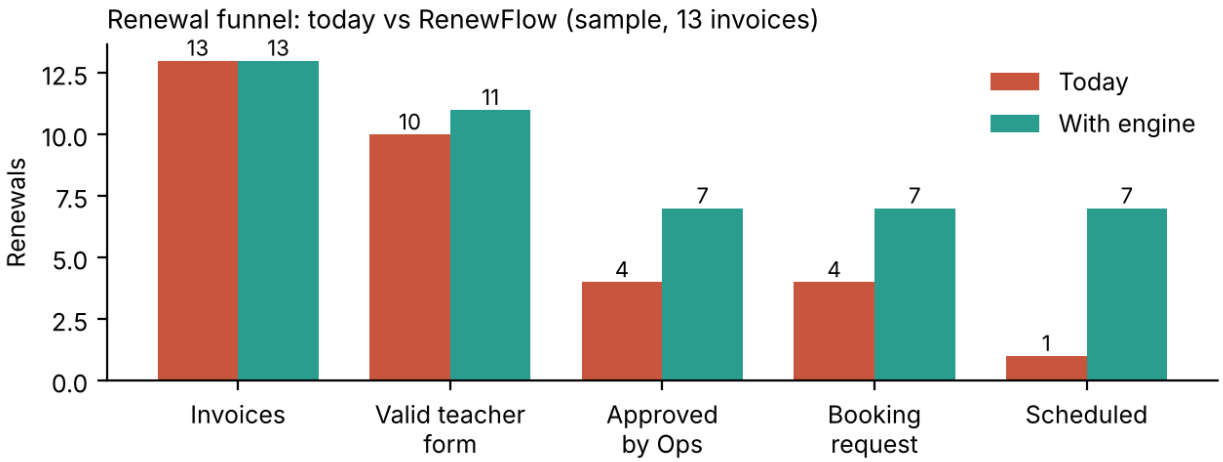


Figure 1. Renewal funnel today versus with the engine (sample data).

Statistical summary (at 9:00 AM, 4 Nov 2026)

Measure	Value
Payment to teacher form	Median 27 min; range about 15 to 50 min (10 of 12 paid invoices); 2 never received a form
Form to Ops approval (4 approved)	0.75 to 2.5 h; median about 1.5 h
Forms still pending (8 of 12)	16 to 48 h; oldest 47.5 h (form 3303); 5 older than a day
Booking requests unscheduled > 12 h	3 of 4; oldest about 64 h (booking 5502)
Paid but unscheduled value	Rs 126,850 across 11 renewals (includes Rs 5,250 collected on one installment)
Sessions scheduled on the one complete renewal	2 of 24 purchased (advisory, cause unknown)

Observed patterns

- **Teachers are fast; Operations is the queue.** A later form (3305) was approved before an earlier clean one (3303): the queue is not first-in-first-out and has no owner.
- **Approval is not progress.** 3 of 4 approved forms stalled one table later (bookings 5502, 5505, 5512).
- **A false 'batch full'.** Booking 5505 failed on capacity although the student already holds one of the six seats.
- **The form retypes what the invoice knows.** Every typo, duplicate and missing form traces to that retyping step.

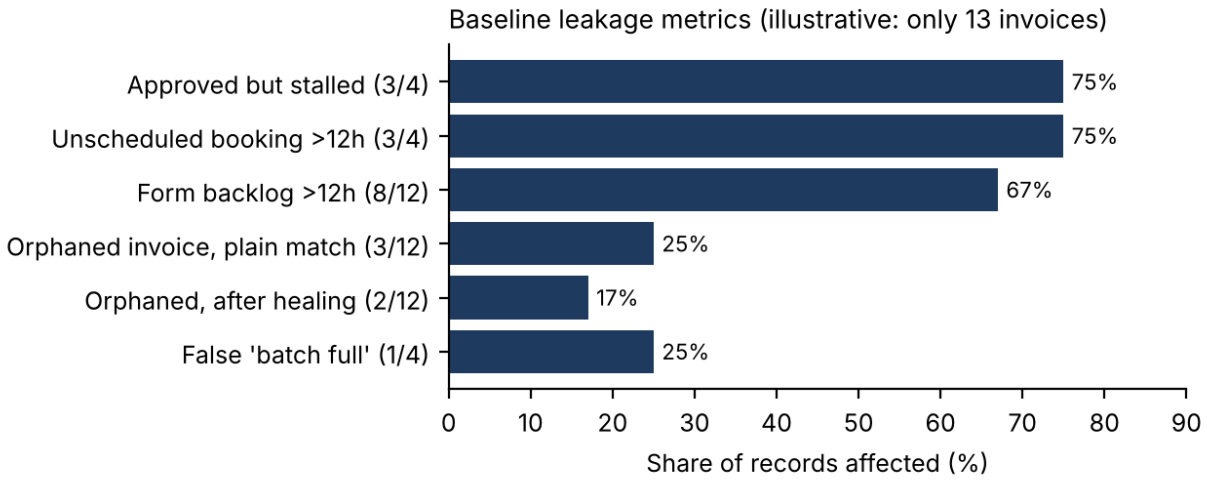


Figure 2. Baseline leakage metrics (percentages are illustrations; the denominators are 4 to 12).

What happened to each invoice

Invoice	Finding	Failure mode
84901	Complete in 1.75 h; only 2 of 24 sessions scheduled	F14 coverage gap
84902	Approved, then idle 64 h in Pending_Batch_Allocation despite valid batch	F11 black hole
84903	Clean same-batch renewal; form pending 47.5 h	F8 unordered queue
84904	Mother paid from another number; good payment, form pending 45 h	F7 payer mismatch
84905	Failed_Batch_Full although the student holds the seat	F12 false batch full
84906 / 84910	Paid (Rs 18,000 / Rs 11,000), no teacher form at all	F1 orphaned invoice
84907	20% discount (Rs 1,900), no approver field	F6 untracked discount
84908	Same form submitted twice, 5 minutes apart	F3 duplicate
84909	First installment Rs 5,250 of 10,500; form claims full payment	F5 installment as full
84911	Payment failed, no payment id, but a form exists (UPI still processing)	F4 form on unpaid invoice
84912	Wants Sat/Sun 11 AM batch that does not exist; Ops approved anyway	F9, F13 no live availability
84913	Form has invoice no. 84931; Cyrillic letters in parent name	F2 typo, F7 look-alike text

8. Data Preprocessing

Step	What the engine does	Sample example
Null handling	Treats NULL and empty as missing	Failed payment id on 84911
Character folding	Maps Cyrillic/Greek look-alikes to Latin	Name on 84913 now matches registry
Masked comparison	Matches visible beginnings and endings of names, phones, emails	Masked parent details
Typo healing	Scores candidate invoices; heals only a single candidate of 0.9 or more	3313: 84931 to 84913 (score 1.0)
Duplicate suppression	Identical student, class, batch, amount, sessions, choice and slot within 24 h: keep earliest; differing forms raise FORM_CONFLICT	3309 dropped
Payment gate	Paid only if status Paid, gateway id and paid-at present, not deleted	84911 blocked, 84909 PAY_PARTIAL
Time normalisation	All times IST; reference time configurable	9:00 AM 4 Nov 2026

Outliers: the 64-hour and 47.5-hour waits are real signals rather than errors, so they are kept and surfaced as breaches. The 5,000-invoice scale test is synthetic (the sample repeated), used only to check performance.

9. Methodology

Why this approach. The data is small, structured and rule-governed (money, identity, capacity, links), so a deterministic engine is more auditable than a model: anyone can ask *why* a renewal was approved and get an exact answer. A language model is used only where the input is language. Before writing code I built a hand-made **golden answer** for all 13 invoices and 8 metrics, and judged the engine against it.

Design principles

- Payment is the trigger and source of truth; no human action starts a renewal.
- Same batch, same teacher and clean money means no human touches it.
- Rules decide money, identity and capacity; AI only reads remarks and writes messages.
- Every exception has a reason code, a named owner role, a timer and a ready draft.
- Safe by design: idempotent, audit-logged, shadow mode first, kill switch.

Questioning the teacher form

Student, class, teacher and amount are already on the invoice; proof of payment is the gateway record; teacher credit is the educator id on the invoice. The form really adds only *intent* (same batch or change) and a remark, and intent can come from the parent in one tap with 'same batch' as the safe default. So the form is removed from the critical path rather than automated.

Techniques, tools and libraries

Layer	Choice	Reason
Engine	Python, pure functions, dictionary lookups	Testable, readable by non-programmers, fast
Text handling	Look-alike folding, masked comparison, edit distance (swap = 1 edit)	Defeats Cyrillic and typo traps
Link healing	Weighted score: user 0.30, student 0.30, teacher 0.10, amount 0.15, near-identical id 0.15; heal only if exactly one candidate scores 0.9 or more	Never heals on weak evidence
Server / UI	FastAPI; plain HTML, CSS, JS; Chart.js stored locally	Three-command run, no build step
AI	Gemini via server-side call; 15 s limit, max 3 concurrent, cached, masked inputs	Language only; falls back to templates
Production design	n8n, WhatsApp Business API, Slack, Metabase on MySQL, Google Sheets override log	No-code, proposed and structure-checked only
Testing	pytest (44 tests), golden answer, shadow report	Prevents silent drift

10. Implementation

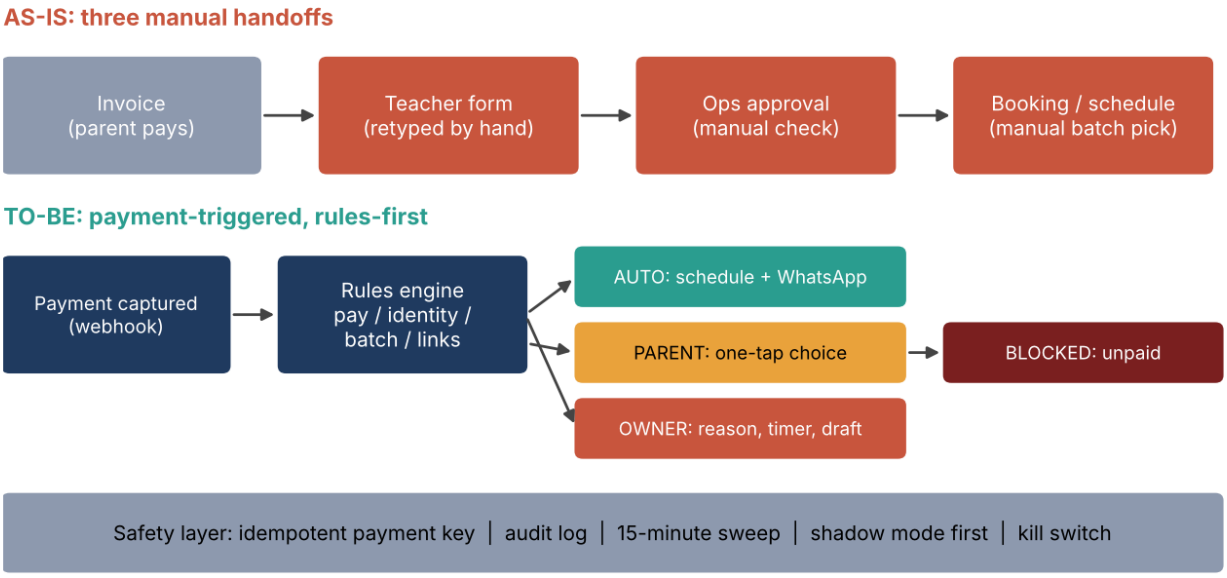


Figure 3. As-is (manual) versus to-be (payment-triggered) workflow.

System architecture

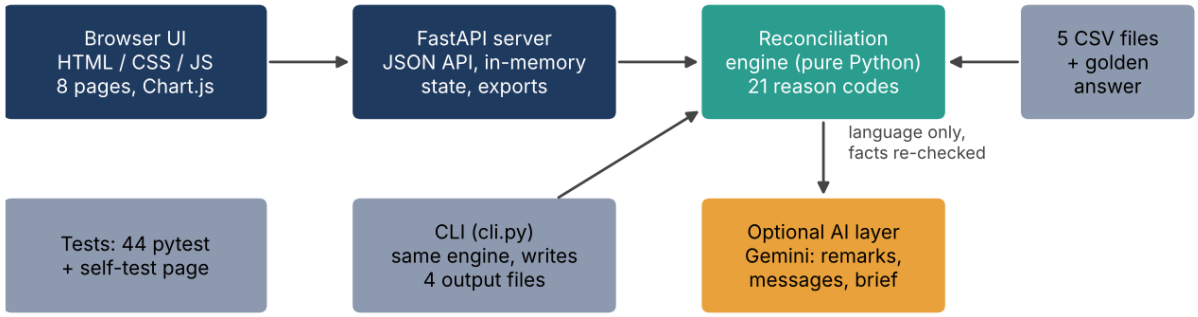


Figure 4. RenewFlow architecture: the engine has no web or AI dependency.

Decision logic

Each renewal receives exactly one of five outcomes, chosen in strict order. The owner is picked from a fixed priority list of reason codes, so the same case always goes to the same role. The sketch below is an illustrative simplification of the rules in this chapter (the full implementation is in the repository).

```
def decide(case):
    if any(c.severity == "block" for c in case.failed):      return "BLOCKED"
    if any(c.severity == "warn" for c in case.failed):
        if case.paid and not case.discount and not case.form \
           and only_issue(case, "LINK_ORPHAN_INVOICE"):        return "FAST_TRACK_PENDING_PARENT"
        return "FLAGGED"
    if case.booking_complete and case.schedule_rows:           return "ALREADY_COMPLETE"
    if case.batch_change and case.matching_batches:            return "FAST_TRACK_PENDING_PARENT"
    return "AUTO_APPROVE"   # notices / info never stop approval

def heal_link(form, invoices):          # weights from the design
    score = lambda i: (.30*(i.user==form.user) + .30*(i.student==form.student)
        + .10*(i.teacher==form.teacher) + .15*(i.net==form.amount)
        + .15*(edit_distance(i.id, form.invoice_id) <= 2))
    hits = [i for i in invoices if score(i) >= 0.9]
    return hits[0] if len(hits) == 1 else None
```

Worked example: how three invoices flow through the engine

Invoice	What the engine sees	Result
84913	Form carries 84931; no such invoice. Candidate 84913 scores 1.0 (user 0.30, student 0.30, teacher 0.10, amount 0.15, near-identical id 0.15). Cyrillic letters in parent name folded to Latin.	Healed, auto-approved; teacher gets a gentle note
84905	Booking marked Failed_Batch_Full; batch is 6 of 6, but the student already holds one of the six seats. Same-batch renewals are seat-neutral.	BATCH_FULL_FALSE_POSITIVE corrected, auto-approved
84911	Status Failed, no gateway payment id, but a form exists. Payment gate fails.	BLOCKED to Finance Ops; parent told not to pay again

Exception routing and AI boundary

Reason code	Owner	Time limit	Action
PAY_NOT_CONFIRMED (block)	Finance Ops	2 h	Re-poll gateway; tell parent not to pay again if debited
PAY_PARTIAL / PAY_AMOUNT_MISMATCH	Finance Ops	4 h	Hold or pro-rata per policy; correction to teacher
DISCOUNT_UNAPPROVED	Ops Lead	4 h	Approve/reject, record approval reference
LINK_ORPHAN_INVOICE	Parent, then Ops agent	4 h	One-tap WhatsApp: same batch or change
BATCH_CHANGE_NO_MATCH	Scheduling Desk	24 h	Open a batch or offer closest alternatives
ID_STUDENT_MISMATCH / ORPHAN_FORM / FORM_CONFLICT	Data Desk	4 to 8 h	Verify which record is wrong

AI is used three times, each checked by something that is not AI: (1) reading teacher remarks, cross-checked against structured fields (raising REMARK_CONTRADICTION); (2) rewriting message templates in English, Hindi or Hinglish, accepted only if every number, date and identifier survives and WhatsApp text stays within 600 characters; (3) the morning brief, with the same number check. Items are labelled AI or Template on screen, and the engine and self-test run fully without AI.

Safety mechanisms

- Idempotency keyed on the gateway payment id; notification log so a message is not repeated every 15 minutes.
- Audit log of approvals, reassignments, notes and messages; proposed rows marked `Regular_Renewal` so one query can reverse them.
- A 15-minute sweep catches missed notifications and over-age cases; a kill switch returns everything to the manual path.

Application screens

The prototype has eight pages: Dashboard, Pipeline (record-chain drawer), Auto Approved (CSV export), Exceptions (by owner, with parent-reply simulation), Messages (edit, translate, regenerate), Process, Shadow and Self Test, and Data and Run. Layouts adapt to phone width, and status is shown in words and symbols, never by colour alone.

[Insert screenshot: Dashboard] Head of Operations view	**[Insert screenshot: Pipeline drawer, invoice 84913]** Healed link and normalised text	**[Insert screenshot: Exceptions page]** Grouped by owner with drafted messages

Figure 5. Screenshots of the working application (to be inserted from the running prototype).

11. Results and Analysis

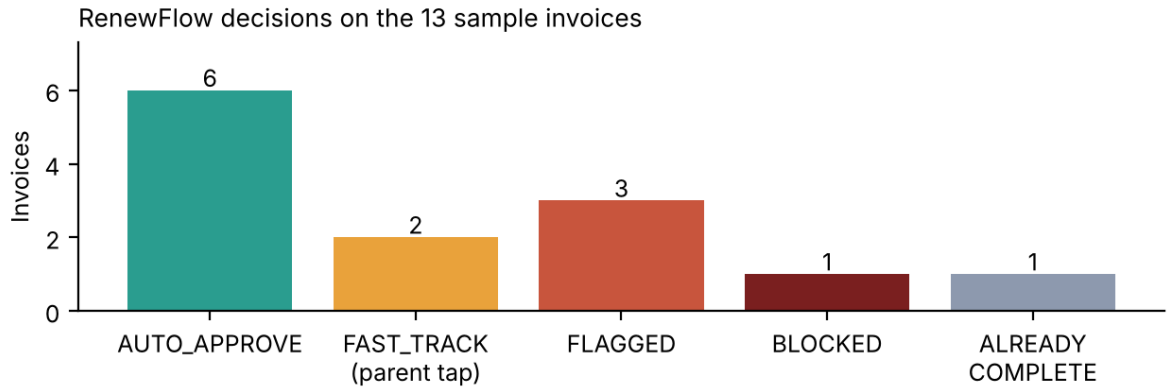


Figure 6. Engine decisions on the sample; all 13 match the hand-built golden answer.

Outcome	Invoices	Owner / note
AUTO_APPROVE (6)	84902, 84903, 84904, 84905, 84908, 84913	144 proposed schedule rows (24 sessions of 50 min each)
FAST_TRACK_PENDING_PARENT (2)	84906, 84910	Paid, no form; one-tap parent choice
FLAGGED (3)	84907, 84909, 84912	Ops Lead (discount), Finance Ops (partial), Scheduling Desk (no batch)
BLOCKED (1)	84911	Finance Ops; parent told not to pay again
ALREADY_COMPLETE (1)	84901	Advisory: 2 of 24 sessions scheduled

Performance and validation

Check	Result
Golden answer	13 of 13 invoices, 8 of 8 metrics matched
Automated tests	44 pass on a clean install (text, linking, payment, messages, AI fallbacks, loading, API, scale)
Scale test	5,005 synthetic invoices in about 2.5 s on the development machine
Touchless rate	6 of 13 (46%); 2 more need only a parent tap; 8 of 13 (62%) need no Operations action; 4 need a person
Messages drafted	30 (parent, teacher, Ops channels); every fact comes from the case record

Shadow-mode comparison with humans

The engine agrees with 3 of the 4 human approvals (forms 3301, 3302, 3305), a 75% agreement rate, so the 95% gate stays closed, which is the correct outcome for four data points. The single disagreement (3312) is the engine being stricter and correct: no batch exists for that request. Across the eight pending forms the engine would remove about 157 hours of waiting and route the other three (about 80 hours) to the right owner at once.

Service levels: targets and sample baselines

Stage	Target	Baseline in sample
Payment to teacher form	2 h today; none once form optional	Median 27 min; 3 paid invoices with no valid form past target
Form to Ops approval	1 h today; under 15 min automated	Median 1.5 h for approved; 11 forms past target
Approval to batch allocation	30 min	3 booking requests unscheduled over 12 h
Allocation to schedule rows	15 min	Not separately observable
Payment to schedule (P90)	Under 4 h clean, under 24 h exceptions	Only 1 complete (1.75 h); 11 paid renewals unscheduled over 4 h

Business impact (assumption-based model)

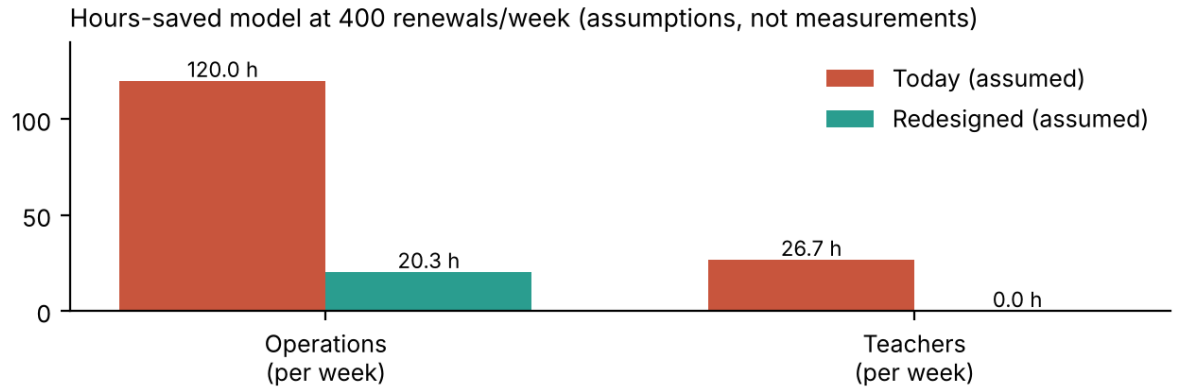


Figure 7. Hours saved: 400 renewals/week; Ops 18 min each today vs target mix (60% auto at 0.5 min, 25% parent tap at 2 min, 15% exceptions at 15 min = 3.05 min average).

Operations falls from about 120 to about 20.3 hours a week (saving about 99.7), and teachers save about 26.7 hours a week of form filling. These are assumptions, to be replaced with real volumes and handling times.

12. Discussion

- **The delay is a process problem, not a people problem.** Teachers act within minutes; waiting accumulates in an unowned queue and in handoffs nobody watches. Automating only the teacher step would have sped up the fast part and left the slow part untouched.
- **Soft versus hard checks matter.** Treating a different payer as a notice (the link is bound to the student id) removes unnecessary hesitation, while failed payments and identity mismatches block hard. A false 'batch full' is a logic error: a renewing student already holds the seat.
- **Ordering by money and age.** The 'Needs action now' list ranks by amount times hours waited, starting with invoice 84906 (Rs 18,000, 41 h), so effort goes where the family and the business are most exposed.
- **Rules own the risky decisions.** Mixing a language model into amounts, identity or seats would make decisions unauditable. Keeping AI to wording, behind a facts check, gives the speed of AI with the traceability of rules.
- **Operations' role shifts** from checking everything to owning exceptions and the parent experience; no one is removed by this project.

Proposed dashboard for the Head of Operations

Six headline numbers (paid but unscheduled Rs 126,850 over 11 renewals; paid with no form over 12 h; forms pending over 12 h with the oldest at 47.5 h; booking requests unscheduled over 12 h; touchless rate 46%; hours saved per week), a five-line morning brief, a today-versus-engine funnel, ageing buckets, a Pareto of reason codes, workload per owner, a ranked action list and service-level panels. The main service level is **payment to schedule**: 90th percentile under 4 h for clean renewals and under 24 h for exceptions.

13. Challenges and Limitations

Challenges encountered

- Traps in the data: swapped digits in a key, invisible Cyrillic characters, masked values, a duplicate form and a failed payment with a form attached.
- Missing facts: no discount approver field, no session history (so Class Gap Days cannot be computed), and no live class-schedule table.
- Policy choices that belong to the business (installments, discount authority), handled by explicit defaults and settings.

What is proven and what is not

Proven	Not proven
Engine reproduces the hand analysis on all 13 invoices; 44 tests pass; scales to about 5,000 invoices in seconds; pages behave correctly in an automated browser simulation; AI path accepts a fact-preserving rewrite, rejects a changed number and falls back on failure (tested with a stand-in service).	Only 13 invoices, so percentages are illustrations, not statistics. AI tested with a stand-in, not a live key; Hindi and Hinglish quality needs human review. Phone layout not checked on real devices. n8n flow and SQL views structure-checked only, not run on live n8n or MySQL. Nothing is sent; approvals and audit log are in memory and lost on restart. Hours saved rests on assumptions.

Open questions to confirm with the Head of Operations

- Installments: hold and flag (default) or schedule the paid share (12 of 24 for invoice 84909)?
- Teacher credit: is it calculated from the form in any way not visible in the data? This must be settled before the form is removed.
- Are sessions scheduled on a rolling basis (explaining 2 of 24), and are seats released at term end?
- Have parents opted in to WhatsApp messages? Are rupee invoices for Gulf and US parents intentional?

14. Conclusion and Recommendations

A parent who pays to renew has done their part. What happens next should not depend on a teacher remembering a form, on Operations reaching a particular row, or on a batch having a free seat the student already holds. RenewFlow shows that clean renewals can be approved in minutes and every other case routed to a named owner with a reason, a timer and a draft.

Recommendations

- **Make the form optional.** Start every renewal from the captured payment; keep a teacher note field and unchanged teacher credit.
- **Ship the cheap fixes first:** an invoice picker instead of a pasted link, a duplicate-submit guard, a seat rule that excludes the renewing student, and a nightly orphaned-invoice alert to Slack.
- **Add an approver and reason code to discounts** on the invoice itself.
- **Give weekends explicit cover:** a named on-call agent, especially for Friday-evening payments for Saturday batches.

30, 60 and 90 day rollout

Stage	Scope	Exit target
Days 1 to 30: shadow	Engine reads only; compare daily with humans; publish baseline SLAs and dashboard; cheap fixes; pilot with 10 teachers and 2 agents	Baseline published; disagreements reviewed
Days 31 to 60: assist	Payment creates the case; form pre-filled and optional; Ops approves the green list in bulk; WhatsApp confirmations to about 50 teachers; Exception Desk with owners and timers	Touchless 40%; no paid invoice unscheduled after 24 h
Days 61 to 90: autopilot	Green-path renewals write to booking and schedule tables; parent one-tap for batch changes; form removed for renewals; all 200+ teachers	Touchless 65% or more; P90 payment to schedule under 4 h

Gates: at least 95% agreement with human decisions and no wrong schedule in the previous two weeks, plus a clean 10% audit sample before autopilot. Roll out by cohort (subject and region), avoid term ends and festival peaks, keep the kill switch ready from day one, and apply minimum-necessary handling to children's data (taking advice on India's DPDP Act).

Possible improvements

- Run on a few weeks of real data in shadow mode and tune rules from every disagreement.
- Write decisions to a persistent staging table; connect a live WhatsApp sandbox; review Hindi and Hinglish output with a native reader.
- Add Class Gap Days measurement and a revenue view per owner once session history is available.

How the work was done

I built this project with the help of an AI assistant (Claude) for analysis, rules, code and writing, and I checked the result against a hand-made golden answer. The prototype can optionally call Google's Gemini model at run time for language tasks.

15. References and Appendix

- Bambinos.live Stage 2 take-home case brief and the five CSV files, official assessment repository (`ops/data`).
- RenewFlow repository: README, `docs/` (case analysis, architecture, SQL views, n8n export), `results/` (sample outputs), `tests/` (44 tests).
- Run: create a virtual environment, install requirements, `uvicorn app.main:app`; tests: `pytest`; CLI: `python cli.py`.

Appendix pages (optional, outside the 12 to 15 page body) follow: reason code reference and glossary.

Appendix A: Reason Code Reference (21 codes)

Code	Sev.	Owner	Limit	Meaning and fix
PAY_NOT_CONFIRMED	block	FINANCE_OPS	2 h	Invoice not Paid. Re-poll gateway; tell parent not to pay again if debited.
PAY_PARTIAL	warn	FINANCE_OPS	4 h	Only part collected. Hold, or schedule paid share if policy enabled; remind parent.
PAY_AMOUNT_MISMATCH	warn	FINANCE_OPS	4 h	Form amount differs from invoice net. Trust invoice; teacher corrects form.
FEE_MISMATCH	warn	FINANCE_OPS	4 h	Invoice not the standard fee for sessions bought. Check price or custom plan.
DISCOUNT_UNAPPROVED	warn	OPS_LEAD	4 h	Discount with no approver. Approve or reject; record reference.
LINK_ORPHAN_INVOICE	warn	PARENT	4 h	Paid, no form. One-tap parent choice; Ops agent after 4 h.
LINK_ID_TYPO_HEALED	notice	NONE	FYI	Form id healed to the single matching invoice. Teacher gets a gentle note.
LINK_DUPLICATE_FORM	notice	NONE	FYI	Duplicate dropped, earliest kept.
ORPHAN_FORM	warn	DATA_DESK	8 h	Form points at a missing invoice that cannot be healed.
FORM_CONFLICT	warn	DATA_DESK	8 h	Multiple forms disagree. Ask teacher which is correct.
ID_PAYER_MISMATCH	notice	NONE	FYI	Payer differs from registered parent; notify registered parent.
ID_STUDENT_MISMATCH	block	DATA_DESK	4 h	User/student ids disagree or no active record.
DATA_TEXT_NORMALIZED	info	NONE	FYI	Look-alike characters normalised; clean at source.
BATCH_CHANGE_NO_MATCH	warn	SCHEDULING_DESK	24 h	No batch with requested days/time and free seat. Open one or offer closest.
BATCH_CHANGE_PARENT_SELECTION	notice	PARENT	24 h	Matching batches exist; parent picks via WhatsApp.
BATCH_RECORD_MISMATCH	warn	SCHEDULING_DESK	8 h	Batch, class or educator differ from active record.
BATCH_SLOT_MISMATCH	warn	SCHEDULING_DESK	8 h	'Same batch' but days or time differ. Ask about a change.
BATCH_FULL_FALSE_POSITIVE	notice	NONE	FYI	Failed as full though the student holds the seat; corrected.
BOOKING_STUCK_RESCUED	notice	NONE	FYI	Idle booking with valid batch allocated automatically.
SCHEDULE_COVERAGE_LOW	info	NONE	FYI	Fewer sessions scheduled than purchased (advisory).
REMARK_CONTRADICTION	warn	DATA_DESK	8 h	Remark contradicts form fields. Teacher confirms.

Appendix B: Glossary

As-of time: reference moment for ages and delays (9:00 AM, 4 Nov 2026). **Batch**: a group of up to six students meeting on set days and times. **Booking request**: table created when Ops approves a form. **Healing**: matching a form to the right invoice despite a typo. **Orphaned invoice**: paid invoice with no form. **Shadow mode**: running the engine without writing, to compare with humans. **Touchless rate**: share of renewals needing no human action.